

01

Code specifications

Cuts applied in Matteo's original code (per channel): MuTau

mutau				
	ttjets	dy	qcd	data
FASE0	1 tau. 1 muon	1 tau. 1 muon	1 tau. 1 muon	1 tau. 1 muon
Doube_medium				
Iso_Mu24	✓	✓	✓	✓
Ele32_WpTight				
Electron_				
lumi filtering			✓	✓
FASE1				
tau id1	>63	>63	>63	>63
tau id2	>7	>7	>7	>7
tau id3	>1	>1	>1	>1
tau pt	>100	>100	>100	>100
mu id	>=3	>=3	>=3	>=3
mu pt	>35	>35	>35	>35
ele id				
ele pt				
delta R	>0.4	>0.4	>0.4	>0.4
eta	<2.4	<2.4	<2.4	<2.4
charge signs	opposite	opposite	same	opposite
scaling factors	✓	✓		
valid jet and MET info	✓	✓		

Selection cuts

1 muon and 1 τ_h (ID+ISO)
 $p_T(\mu) > 35 \text{ GeV}, p_T(\tau_h) > 100 \text{ GeV}$
 Opposite sign (same-sign for QCD)
 At least two protons in PPS

Cuts explained in the analysis
 note
 proton requirements are applied
 after introducing pileup protons

2 Signal and background samples

This study considers the 2018 data taking period. For this reason, samples are simulated taking in account the behaviour of the detector configuration during this year.

2.1 Signal samples

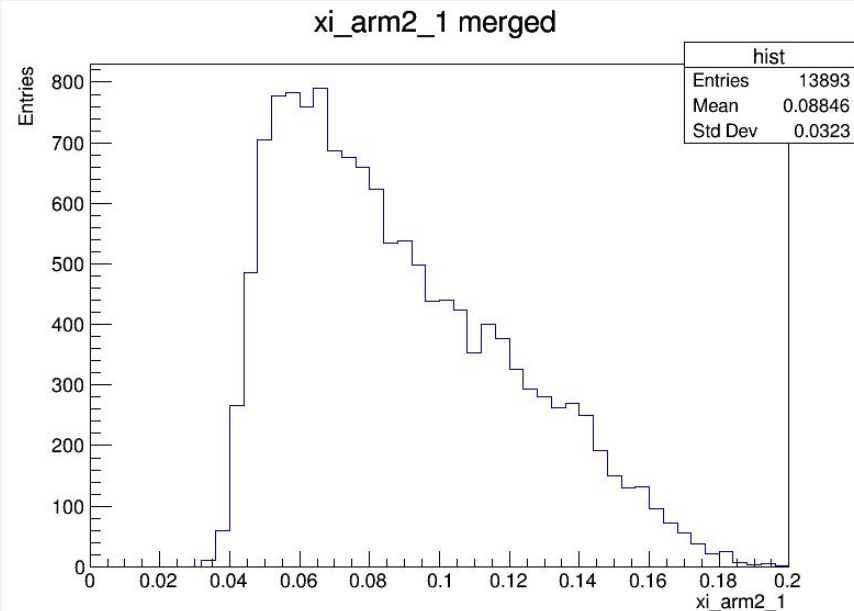
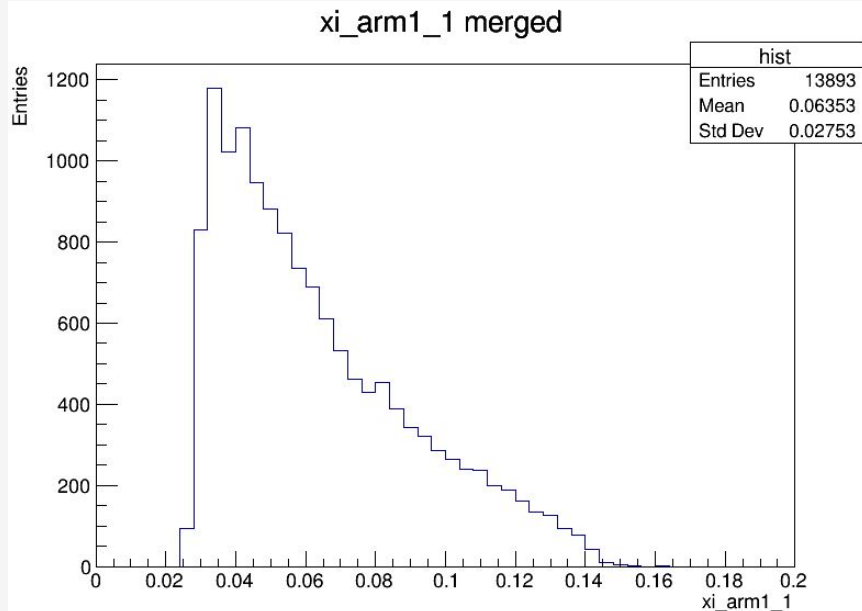
In this analysis, we are interested in testing the SM predictions and BSM physics models. The signal sample is generated using MadGraph5_aMC@NLO v3.5.2 [8–10], which includes recent charge form factors of photon fluxes and hadronic survival probabilities [11]. Two BSM models are considered: high mass resonances and EFT modifications of the $\gamma\tau$ coupling. The BSM model¹ is implemented using the SMEFTsim package [12], with $\alpha^{-1} = 137$.

- Reads muon scale factors
 - loads proton acceptance correction and systematic uncertainties
 - Muon corrections: Trigger, Id, ISO, Reco
 - Applies tau/muon ID cuts + η
 - Normalization with BSM weights in proper branches
- (proton cuts expanded in next slide)

Pronton ξ AFTER sinal.cpp

Proton related cuts:

- 2 reconstructed protons
- fiducial acceptance cuts - base on ξ and θ_x
- period dependent geometric cuts
- radiation damage corrections - weight factors applied from proper root files according to run period
- ξ systematic variations - evaluation of systematic uncertainties -> produces up and down shifted ξ values per arm



What changed in the processing

- Processing all samples in the same way (DY, ttjets, QCD, Data) was changed to:
 - DY and ttjets: simulated samples; they go through 2 phases of cuts and pileup protons are added at the end of this process.
 - Data and QCD: real data; instead of “losing” the proton related variables, they are saved from the first step; there is no need to add pileup protons

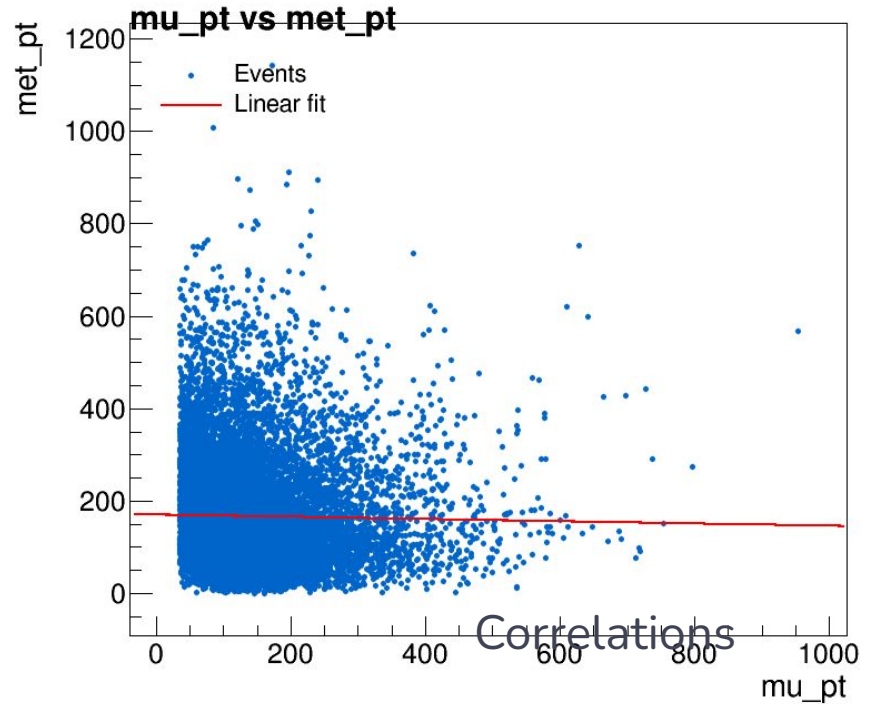
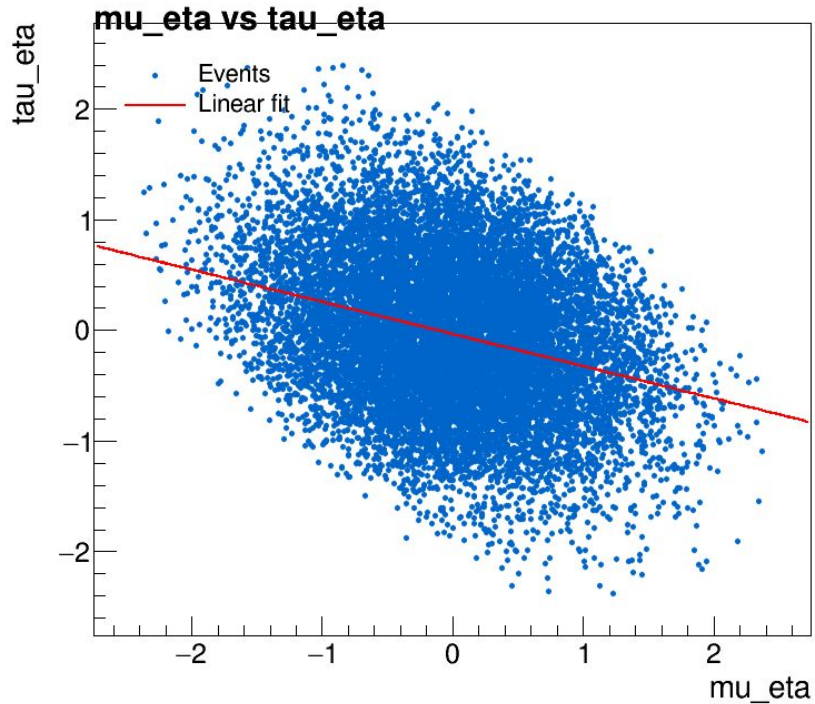
02

Variable

plots

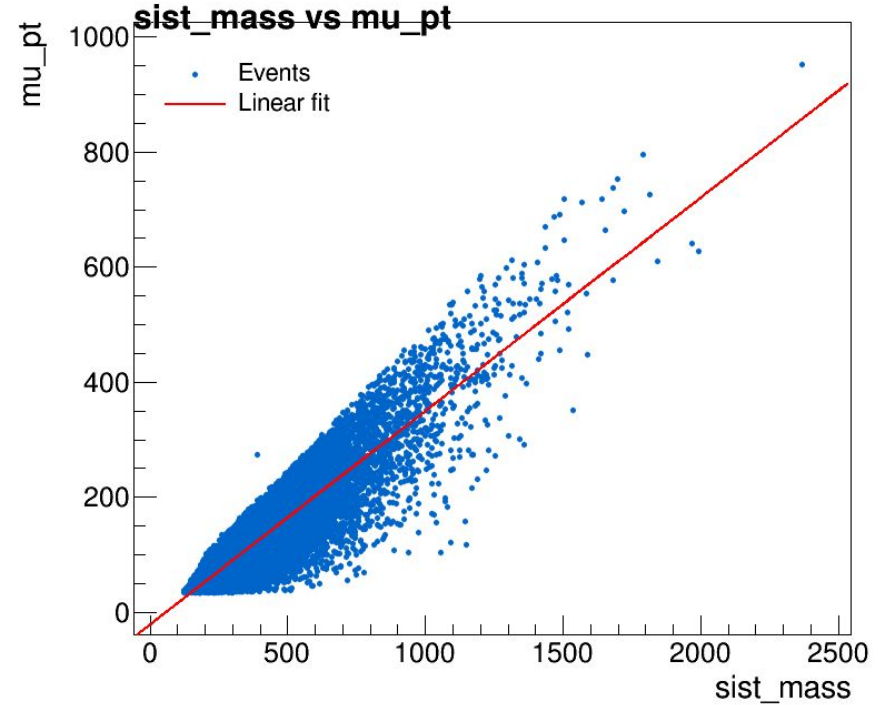
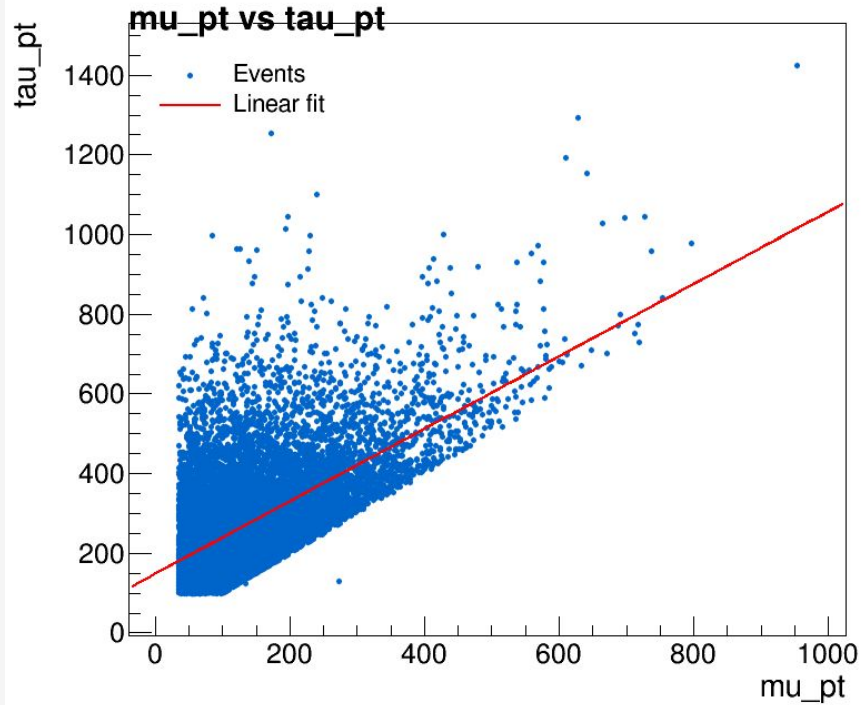
Correlations

Correlation between variables: signal

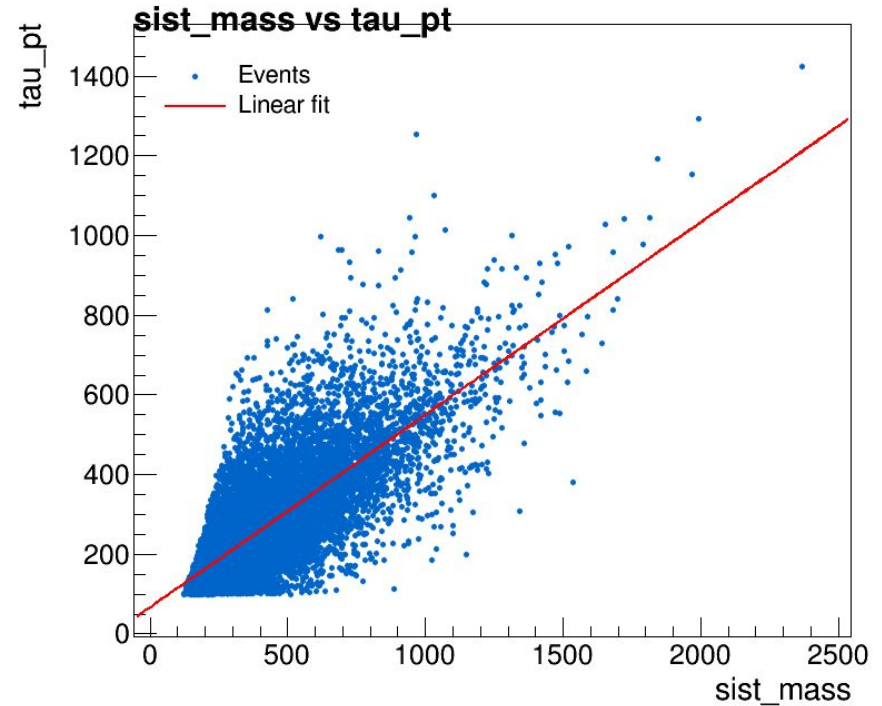
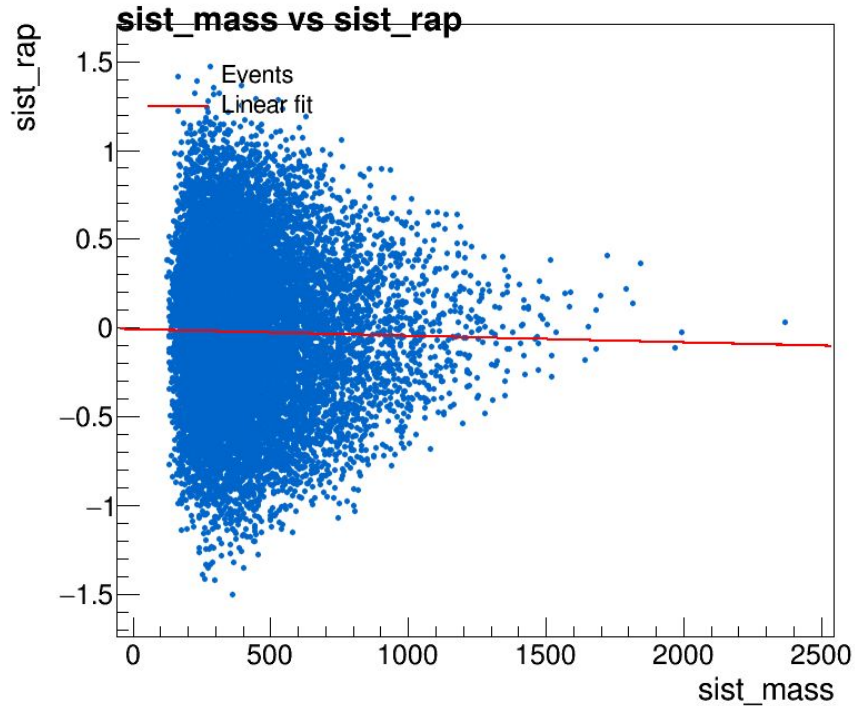


Correlations

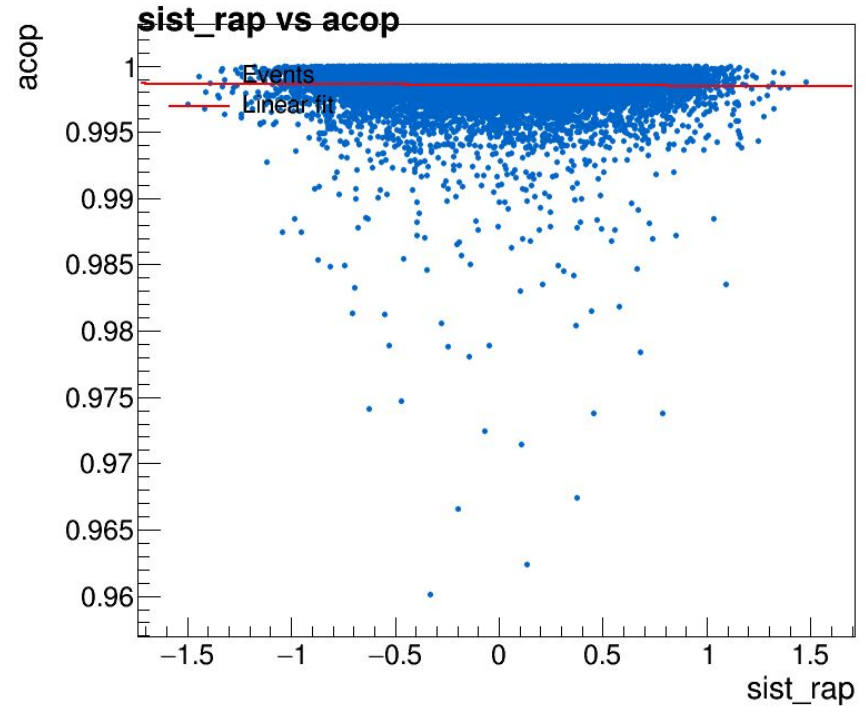
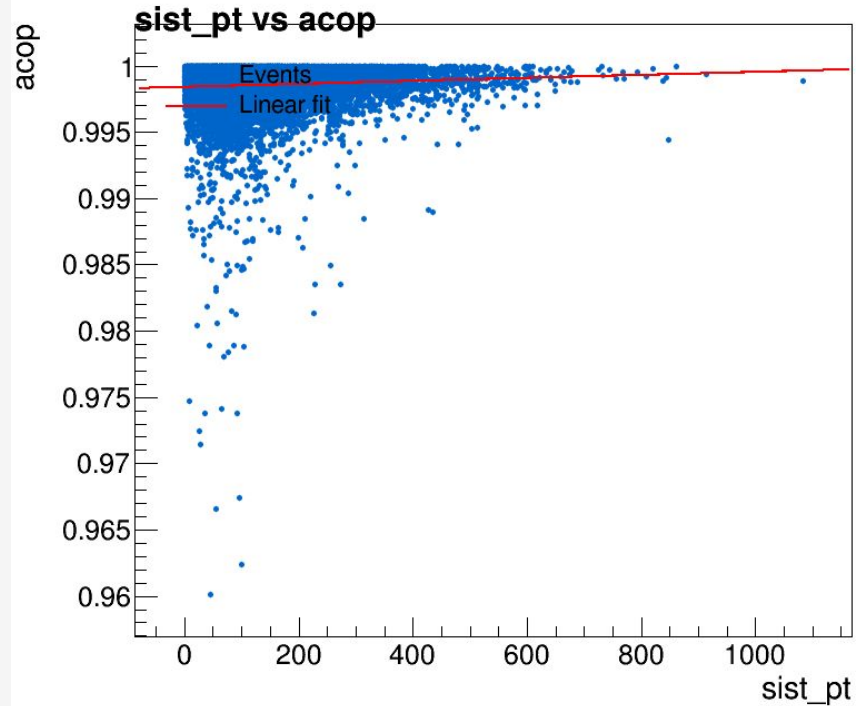
Correlation between variables: signal



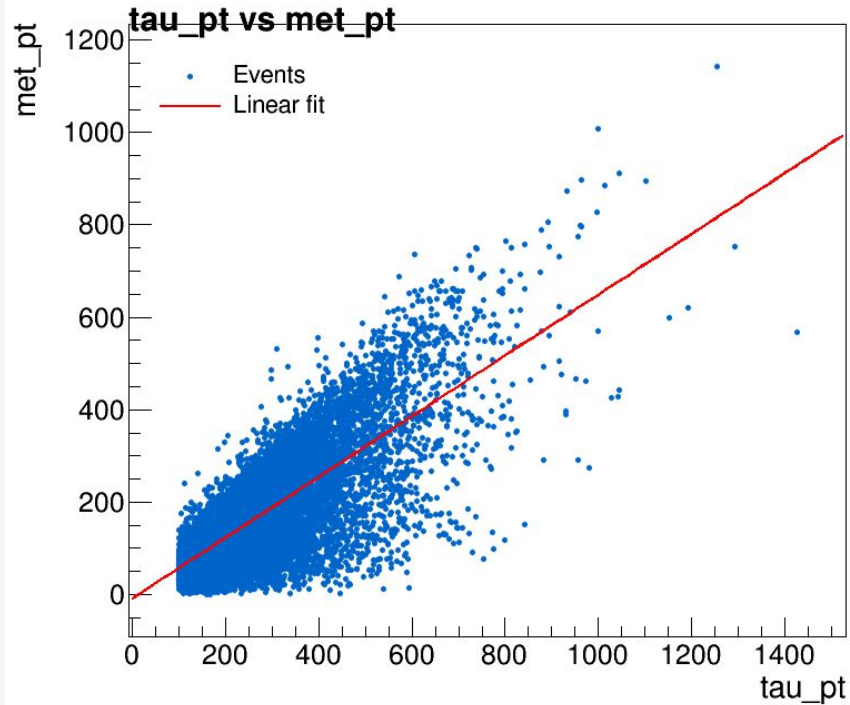
Correlation between variables: signal



Correlation between variables: signal



Correlation between variables: signal



02

Variable plots

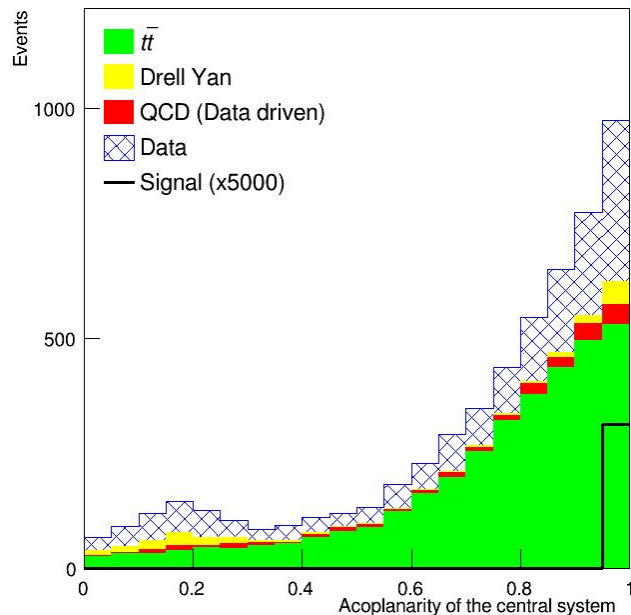
Input for the BDT

Input for the BDT; signal magnified for visualization

CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

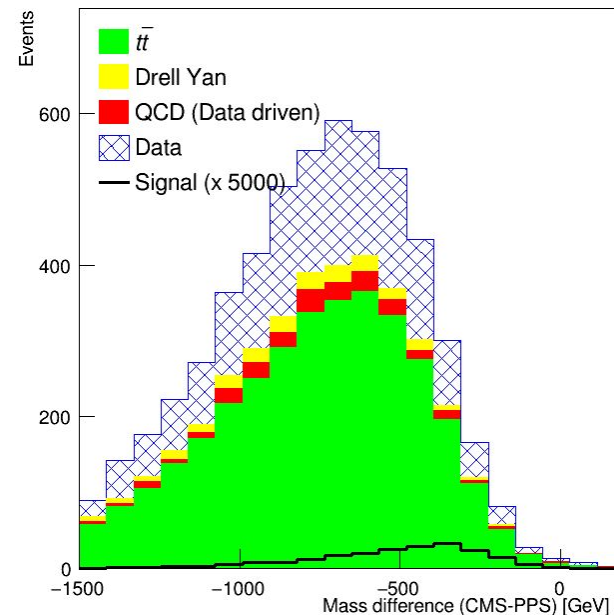
Preliminary



CMS-TOTEM

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Preliminary

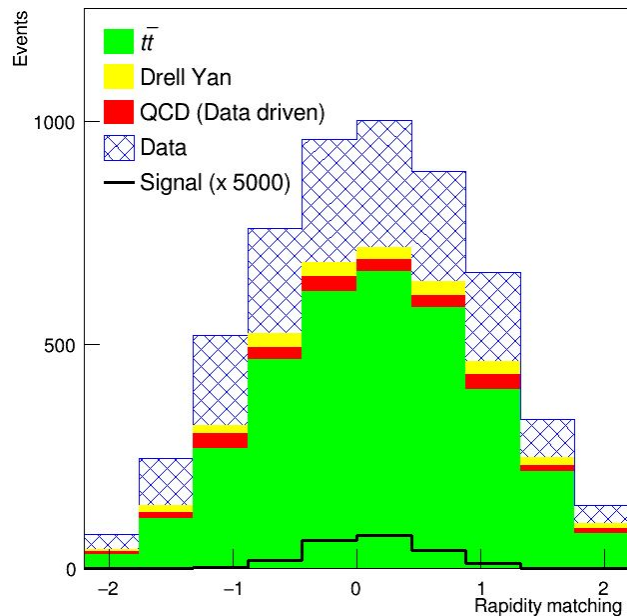


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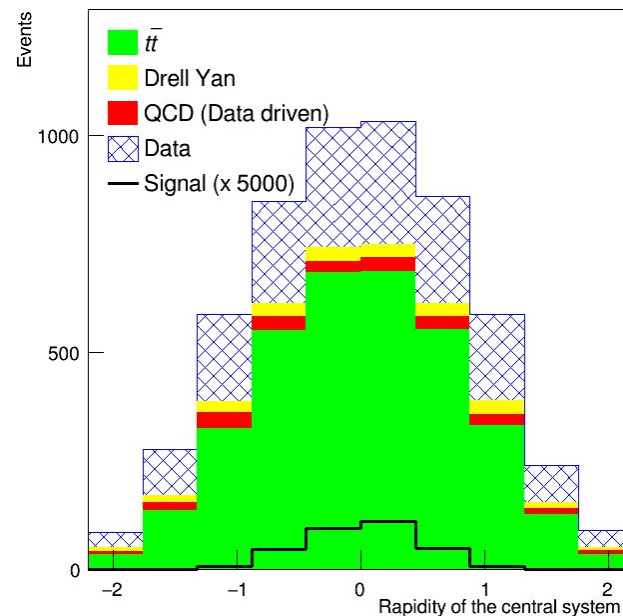
Preliminary



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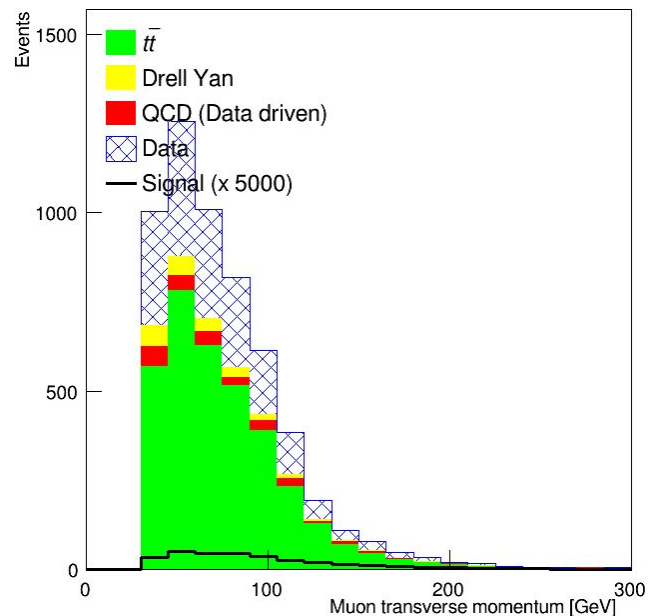


Input for the BDT; signal magnified for visualization

CMS-TOTEM

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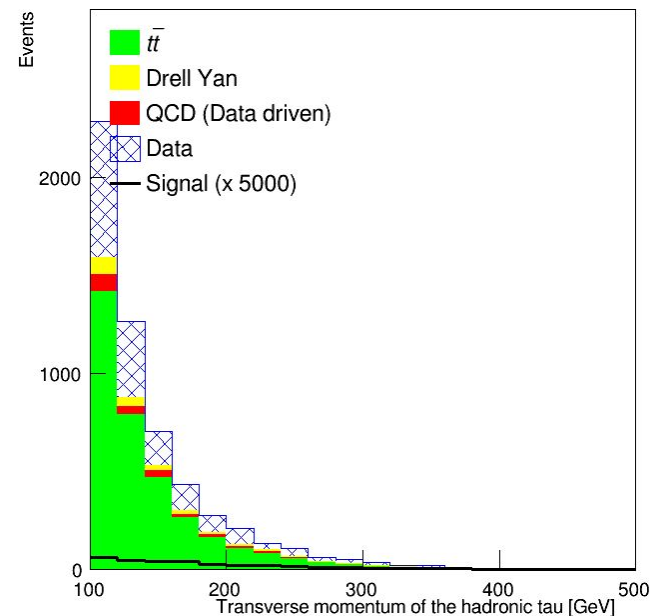
Preliminary



CMS-TOTEM

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Preliminary

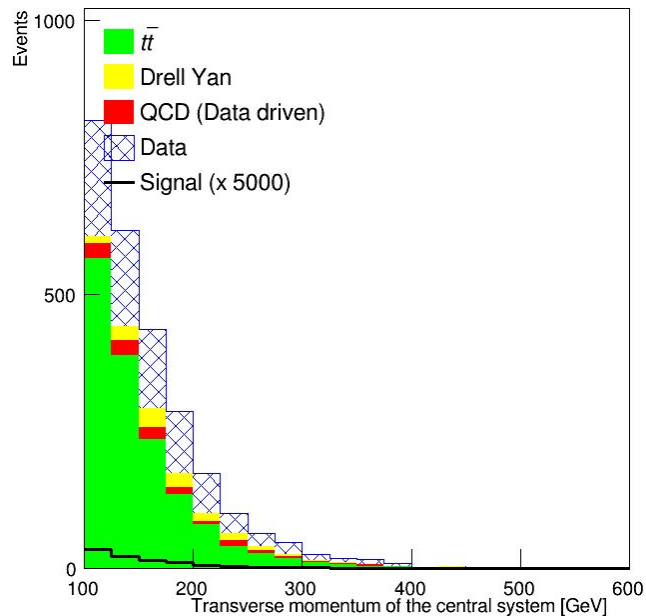


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CMS-TOTEM

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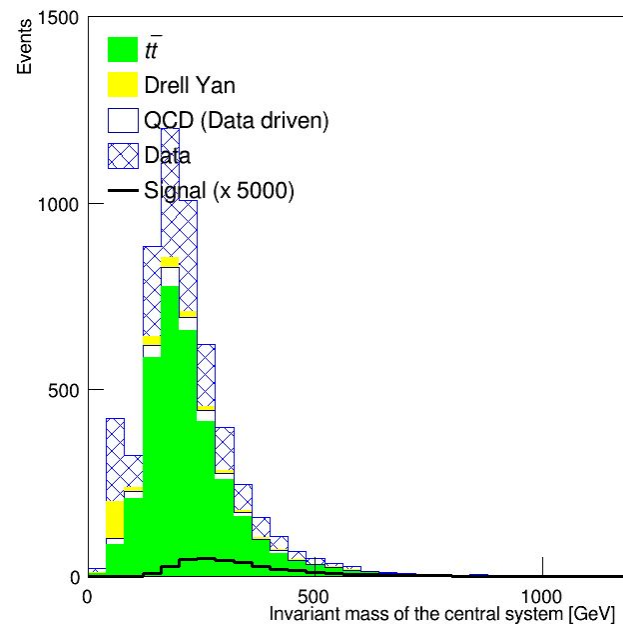
Preliminary



CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

Preliminary



02

Variable plots

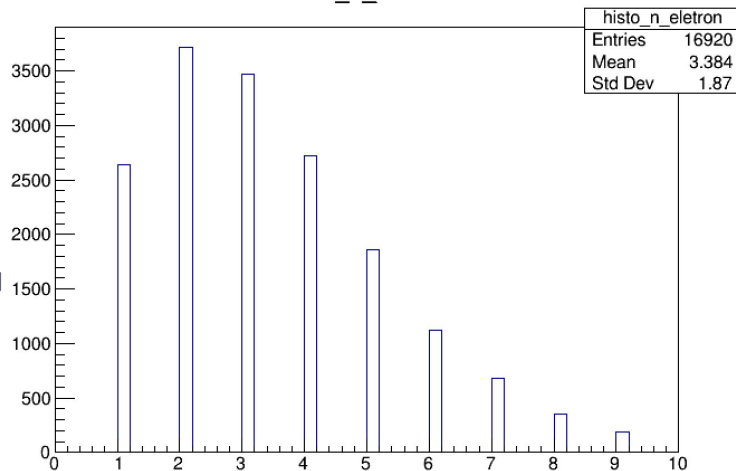
Signal

Signal plots
produced by
sinal.cpp code
mutau channel

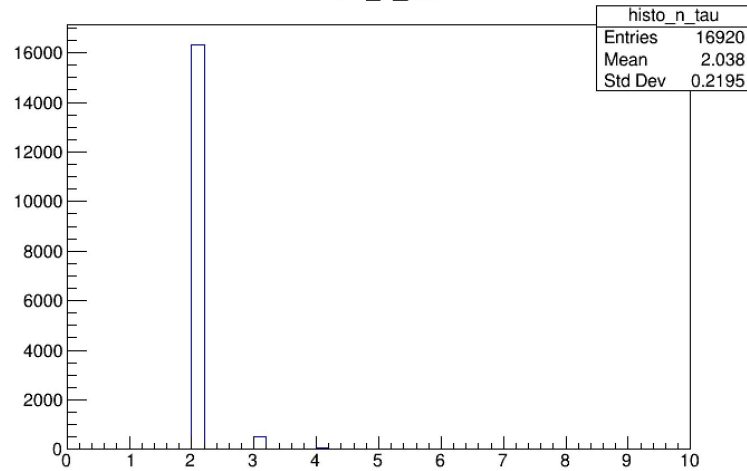
note: where is written e
or *electron* it should read
muon

- μ pT > 35
- tau pT > 100
- $\eta < 2.4$ for both
- opposite sign candidates
- at least one proton per pps arm

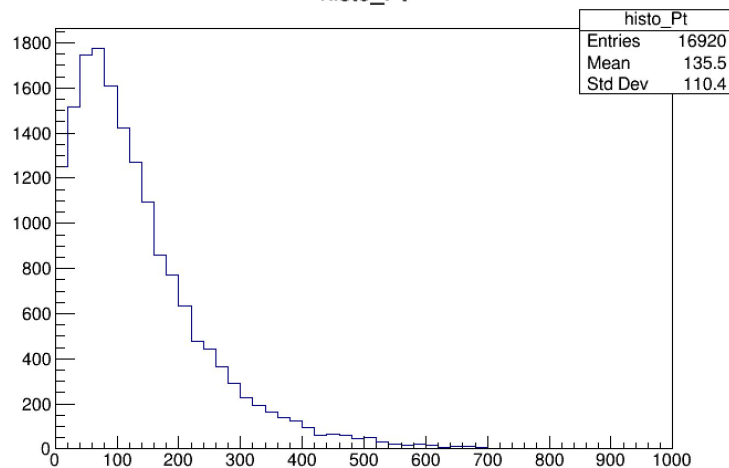
histo_n_eletron



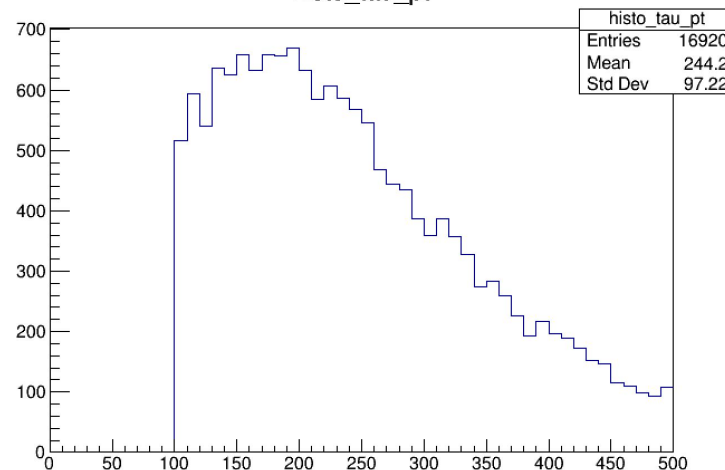
histo_n_tau



histo_Pt



histo_tau_pt

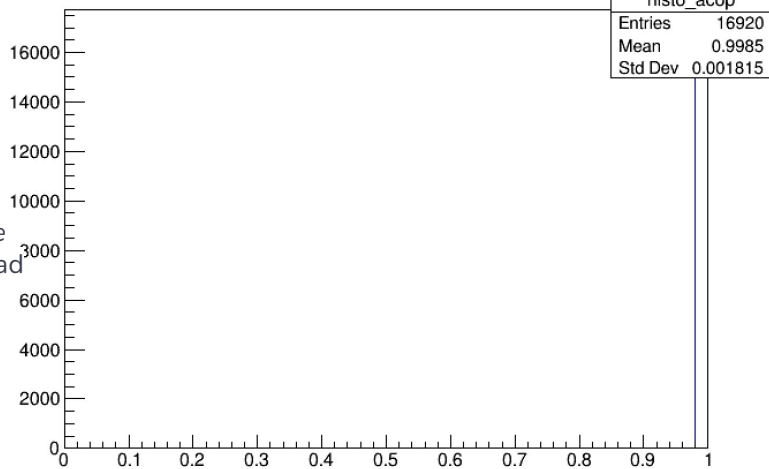


Signal plots
produced by
sinal.cpp code
mutau channel

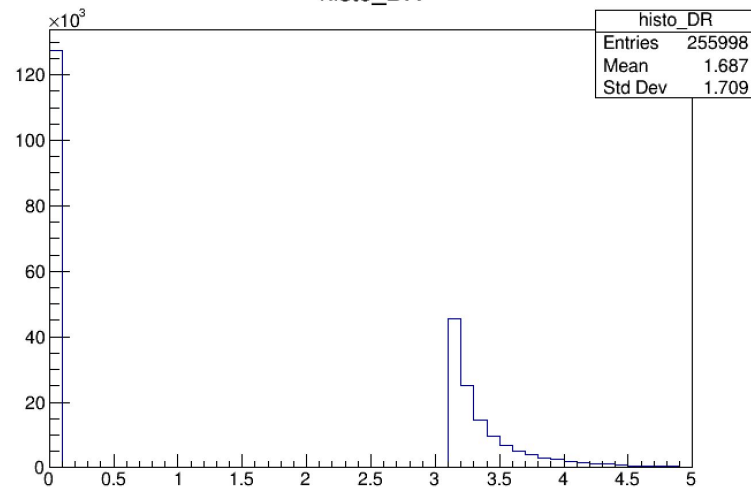
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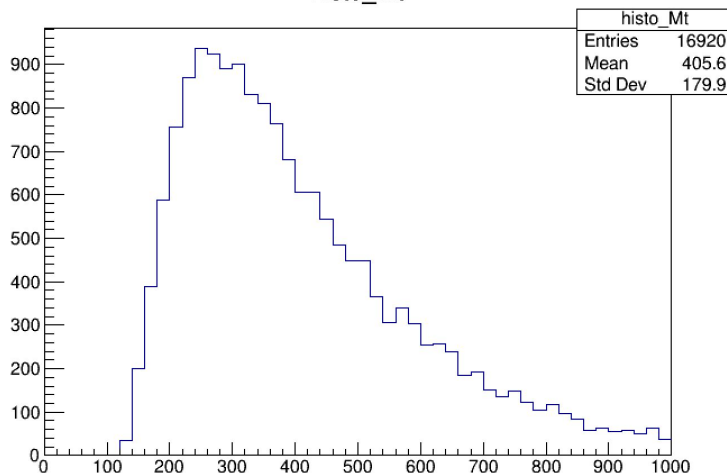
histo_acop



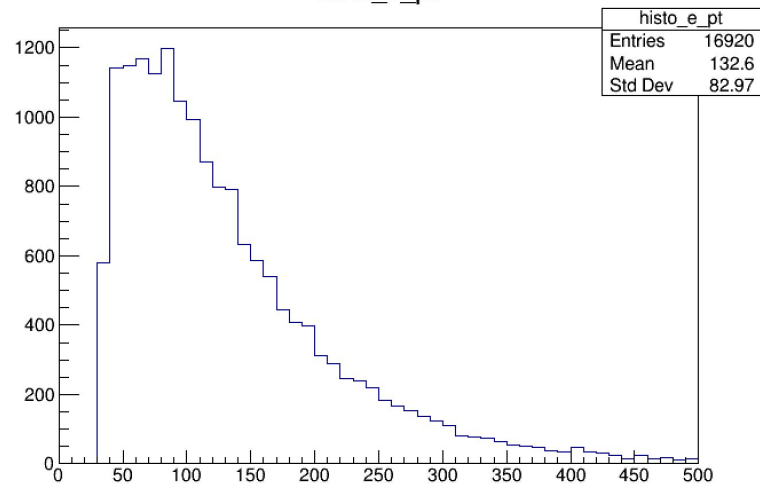
histo_DR



histo_Mt



histo_e_pt



03

Writing

Progress:

Introduction

- Using Matteo's analysis note as starting point
- Precision tau physics -> for descriptions and definitions

Analysis

- data analysis in high energy physics for definitions such as multivariate classification, boosted decision trees, etc
- explanation of the processing applied to the samples for each channel